

HAMZA HAFEEZ

BACKEND + AI SYSTEMS ENGINEER

Lahore, Pakistan (Remote-ready, open to relocate) | hmza.hb82@gmail.com | hamza-hafeez.site

Github: github.com/hmza-hb

Linkedin: linkedin.com/in/hmza-hb

PROFILE

Software Engineer building production AI systems, secure applications, and scalable backend infrastructure. Experienced in architecture, technical scoping, client delivery, and end-to-end product development. Founder of two live products with independent AI research.

EXPERIENCE & DEVELOPMENTS

Software Engineer, Upvista Digital

Jan 2025 - Present

- Architected and delivered full-stack applications for clients across 6 countries, owning technical discovery, system design, and production delivery.
- Built scalable backend systems using Go/FastAPI with PostgreSQL, Redis, Docker, and cloud infrastructure for multi-tenant workloads.
- Designed secure APIs, data architectures, and client integrations while serving as the primary technical point of contact from requirements to deployment.

Cortex: Vulnerability Analysis Platform

Github: github.com/Cortex-EDR

Web: cortex-edr.com

TypeScript • Go • LangGraph • OSV.dev • Multi-Agent AI • Semgrep

- Engineered a 7-agent security analysis platform with an orchestrator based on the PFC function.
- Developed autonomous security agents for vulnerability discovery, exploitability analysis, evidence generation, and reporting.
- Built isolated sandbox environments for attack simulation and vulnerability validation.
- Developed Cortex Attack, an npm-distributed reconnaissance CLI for automated security workflows.

Histeeria: Trust Infrastructure for AI

GitHub: github.com/histeeria

Web: histeeria.com

Go • typescript • FastAPI • PostgreSQL • Redis • RAG • LLM APIs

- Built end-to-end AI agent monitoring SaaS: SDK ingest → async eval pipeline → Next.js dashboard with alerts, reports, and public agent profiles.
- Shipped zero-dependency client SDKs on PyPI and npm (histeeria)
- Built **FastAPI backend** with SQLAlchemy 2, multi-tenant schema.
- Implemented 3-tier LLM evaluation pipeline: rule-based heuristics → OpenAI judge (gpt-5.5) → adjudicator (Fable 5) scoring decisions across 8 judgment dimensions.

Anya: Social Robot (software + hardware)

GitHub: github.com/hmza-hb/Anya

Tsx • Py • NATS • WebSockets • CV • TTS/STT • Multi-Model • Memory Systems

- Built a multimodal AI system combining perception, memory, cognition, and expressive behavior.
- Designed an event-driven NATS architecture for real-time AI subsystem communication.
- Developed agentic workflows with contextual memory, perception pipelines, and adaptive responses.

SKILLS

Languages: Go, TypeScript, Python, SQL, C#.

Frameworks: Next.js, Gin, FastAPI, LangChain.

Infra: Docker, Terraform, NATS, Redis, Kafka, Linux, CI/CD, Distributed systems.

AI/LLM: Multi-Agent Systems, RAG, LLM APIs, LangGraph, LLM Evaluation Pipelines.

Security: OWASP, SAST, Semgrep, CVE Analysis, Secure API Design, Basic Cryptography.

Delivery: Technical Scoping, Customer Communication, Architecture.

RESEARCH

Published independent research proposing a prefrontal cortex-inspired architecture for multi-agent AI orchestration and cognitive system design.

Article: github.com/hmza-hb/Project-Cortex

EDUCATION

National University of Modern Languages

2023-2027

Computer Science, Won 5 hackathons